

2021 AIME II Problems/Problem 7

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Problem

Let a , b , c , and d be real numbers that satisfy the system of equations

$$\begin{aligned}a + b &= -3, \\ab + bc + ca &= -4, \\abc + bcd + cda + dab &= 14, \\abcd &= 30.\end{aligned}$$

There exist relatively prime positive integers m and n such that

$$a^2 + b^2 + c^2 + d^2 = \frac{m}{n}.$$

Find $m + n$.

Solution 1

From the fourth equation we get $d = \frac{30}{abc}$. Substitute this into the third equation and you get

$$abc + \frac{30(ab + bc + ca)}{abc} = abc - \frac{120}{abc} = 14. \text{ Hence } (abc)^2 - 14(abc) - 120 = 0. \text{ Solving, we get } abc = -6 \text{ or } abc = 20. \text{ From the first and second equation, we get}$$

$ab + bc + ca = ab - 3c = -4 \implies ab = 3c - 4$. If $abc = -6$, substituting we get $c(3c - 4) = -6$. If you try solving this you see that this does not have real solutions in c , so abc must be 20. So $d = \frac{3}{2}$. Since

$c(3c - 4) = 20$, $c = -2$ or $c = \frac{10}{3}$. If $c = \frac{10}{3}$, then the system $a + b = -3$ and $ab = 6$ does not give you real solutions. So $c = -2$. Since you already know $d = \frac{3}{2}$ and $c = -2$, so you can solve for a and b pretty easily and

see that $a^2 + b^2 + c^2 + d^2 = \frac{141}{4}$. So the answer is 145.

~math31415926535 ~minor edit by Mathkiddie

Solution 2

Note that $ab + bc + ca = -4$ can be rewritten as $ab + c(a + b) = -4$. Hence, $ab = 3c - 4$.

Rewriting $abc + bcd + cda + dab = 14$, we get $ab(c + d) + cd(a + b) = 14$. Substitute $ab = 3c - 4$ and solving, we get

$$3c^2 - 4c - 4d - 14 = 0.$$

We refer to this as Equation 1.

Note that $abcd = 30$ gives $(3c - 4)cd = 30$. So, $3c^2d - 4cd = 30$, which implies $d(3c^2 - 4c) = 30$ or

$$3c^2 - 4c = \frac{30}{d}.$$

We refer to this as Equation 2.

Substituting Equation 2 into Equation 1 gives, $\frac{30}{d} - 4d - 14 = 0$.

Solving this quadratic yields that $d \in \left\{-5, \frac{3}{2}\right\}$.

Now we just try these two cases:

For $d = \frac{3}{2}$ substituting in Equation 1 gives a quadratic in c which has roots $c \in \left\{\frac{10}{3}, -2\right\}$.

Again trying cases, by letting $c = -2$, we get $ab = 3c - 4$, Hence $ab = -10$. We know that $a + b = -3$, Solving these we get $a = -5, b = 2$ or $a = 2, b = -5$ (doesn't matter due to symmetry in a, b).

So, this case yields solutions $(a, b, c, d) = \left(-5, 2, -2, \frac{3}{2}\right)$.

Similarly trying other three cases, we get no more solutions, Hence this is the solution for (a, b, c, d) .

$$\text{Finally, } a^2 + b^2 + c^2 + d^2 = 25 + 4 + 4 + \frac{9}{4} = \frac{141}{4} = \frac{m}{n}.$$

$$\text{Therefore, } m + n = 141 + 4 = \boxed{145}.$$

~Arnav Nigam

Solution 3

For simplicity purposes, we number the given equations (1), (2), (3), and (4), in that order.

Rearranging (2) and solving for c , we have

$$\begin{aligned} ab + (a + b)c &= -4 \\ ab - 3c &= -4 \\ c &= \frac{ab + 4}{3}. \end{aligned} \tag{5}$$

Substituting (5) into (4) and solving for d , we get

$$ab \left(\frac{ab + 4}{3} \right) d = 30$$

$$c = \left(\frac{ab+4}{3} \right)^2 - \frac{90}{ab(ab+4)}$$

$$d = \frac{90}{ab(ab+4)}. \quad (6)$$

Substituting (5) and (6) into (3) and simplifying, we rewrite the left side of (3) in terms of a and b only:

$$\begin{aligned} ab \left[\frac{ab+4}{3} \right] + b \left[\frac{ab+4}{3} \right] \left[\frac{90}{ab(ab+4)} \right] + \left[\frac{ab+4}{3} \right] \left[\frac{90}{ab(ab+4)} \right] a + \left[\frac{90}{ab(ab+4)} \right] ab &= 14 \\ ab \left[\frac{ab+4}{3} \right] + \underbrace{\frac{30}{a} + \frac{30}{b}}_{\text{Group them.}} + \frac{90}{ab+4} &= 14 \\ ab \left[\frac{ab+4}{3} \right] + \frac{30(a+b)}{ab} + \frac{90}{ab+4} &= 14 \\ ab \left[\frac{ab+4}{3} \right] + \underbrace{\frac{-90}{ab} + \frac{90}{ab+4}}_{\text{Group them.}} &= 14 \\ ab \left[\frac{ab+4}{3} \right] - \frac{360}{ab(ab+4)} &= 14. \end{aligned}$$

Let $t = ab(ab+4)$, from which

$$\frac{t}{3} - \frac{360}{t} = 14.$$

Multiplying both sides by $3t$, rearranging, and factoring give $(t+18)(t-60) = 0$. Substituting back and completing the squares produce

$$\begin{aligned} [ab(ab+4)+18][ab(ab+4)-60] &= 0 \\ [(ab)^2+4ab+18][(ab)^2+4ab-60] &= 0 \\ \underbrace{[(ab+2)^2+14]}_{ab+2=\pm\sqrt{14}i \implies ab \notin \mathbb{R}} \underbrace{[(ab+2)^2-64]}_{ab+2=\pm 8} &= 0 \end{aligned}$$

$$ab = 6, -10.$$

If $ab = 6$, then combining this with (1), we know that a and b are the solutions of the quadratic $x^2 + 3x + 6 = 0$. Since the discriminant is negative, neither a nor b is a real number.

If $ab = -10$, then combining this with (1), we know that a and b are the solutions of the quadratic $x^2 + 3x - 10 = 0$, or $(x+5)(x-2) = 0$, from which $\{a, b\} = \{-5, 2\}$. Substituting $ab = -10$ into (5) and (6), we obtain $c = -2$ and $d = \frac{3}{2}$, respectively. Together, we have

$$a^2 + b^2 + c^2 + d^2 = \frac{141}{4},$$

so the answer is $141 + 4 = \boxed{145}$.

~MRENTHUSIASM

Solution 4 (Way Too Long)

Let the four equations from top to bottom be listed 1 through 4 respectively. We factor equation 3 like so:

$$abc + d(ab + bc + ca) = 14$$

Then we plug in equation 2 to receive $abc - 4d = 14$. By equation 4 we get $abc = \frac{30}{d}$. Plugging in, we get $\frac{30}{d} - 4d = 14$. Multiply by d on both sides to get the quadratic equation $4d^2 + 14d - 30 = 0$. Solving using the quadratic equation, we receive $d = \frac{3}{2}, d = -5$. So, we have to test which one is correct. We repeat a similar process as we did above for equations 1 and 2. We factor equation 2 to get

$$ab + c(a + b) = -4$$

After plugging in equation 1, we get $ab - 3c = -4$. Now we convert it into a quadratic to receive $3c^2 - 4c - abc = 0$. The value of abc will depend on d . So we obtain the discriminant $16 + 12abc$. Let $d = -5$. Then $abc = \frac{30}{-5}$, so $abc = -6$, discriminant is $16 - 72$, which makes this a dead end. Thus $d = \frac{3}{2}$. For $d = \frac{3}{2}$, making $abc = 20$. This means the discriminant is just 256 , so we obtain two values for c as well. We get either $c = \frac{10}{3}$ or $c = -2$. So, we must AGAIN test which one is correct. We know $ab = 3c - 4$, and $a + b = -3$, so we use these values for testing. Let $c = \frac{10}{3}$. Then $ab = 6$, so $a = \frac{6}{b}$. We thus get $\frac{6}{b} + b = -3$, which leads to the quadratic $b^2 + 3b + 6$. The discriminant for this is $9 - 24$. That means this value of c is wrong, so $c = -2$. Thus we get polynomial $b^2 + 3b - 10$. The discriminant this time is 49 , so we get two values for b . Through simple inspection, you may see they are interchangeable, as if you take the value $b = 2$, you get $a = -5$. If you take the value $b = -5$, you get $a = 2$. So it doesn't matter. That means the sum of all their squares is

$$\frac{9}{4} + 4 + 4 + 25 = \frac{141}{4},$$

so the answer is $141 + 4 = \boxed{145}$.

~amcrunner

Solution 5

Let the four equations from top to bottom be listed (1) through (4) respectively. Multiplying both sides of (3) by d and factoring some terms gives us $abcd + d^2(ab + ac + bc) = 14d$. Substituting using equations (4) and (2) gives us $30 - 4d^2 = 14d$, and solving gives us $d = -5$ or $d = \frac{3}{2}$. Plugging this back into (3) gives us $abc + d(ab + ac + bc) = abc + (-5)(-4) = abc + 20 = 14$, or using the other solution for d gives us $abc - 6 = 14$. Solving both of these equations gives us $abc = -6$ when $d = -5$ and $abc = 20$ when $d = \frac{3}{2}$.

Multiplying both sides of (2) by c and factoring some terms gives us $abc + c^2(a + b) = abc - 3c^2 = -4c$. Testing $abc = -6$ will give us an imaginary solution for c , so therefore $abc = 20$ and $d = \frac{3}{2}$. This gets us

$20 - 3c^2 = -4c$. Solving for c gives us $c = \frac{3}{10}$ or $c = -2$. With a bit of testing, we can see that the correct value of c is $c = -2$. Now we know $a + b = -3$ and $ab + bc + ca = ab + c(a + b) = ab + 6 = -4$, $ab = -10$, and it is obvious that $a = -5$ and $b = 2$ or the other way around, and therefore,

$$a^2 + b^2 + c^2 + d^2 = 25 + 4 + 4 + \frac{9}{4} = \frac{141}{4}, \text{ giving us the answer } 141 + 4 = \boxed{145}.$$

~hihithere there minor edit by ~Gustyrustyp

Video Solution

<https://www.youtube.com/watch?v=2rrX1G7iZqg>

Video Solution by Interstigation

<https://youtu.be/fGgbCglHRHM>

~Interstigation

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